



CLDV7111 POE PART 1

Student Name: Koketso Masasanya

Student Number: S10225793

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Koketso Masasanya

ST10225793@RCCONNECT.EDU.ZA

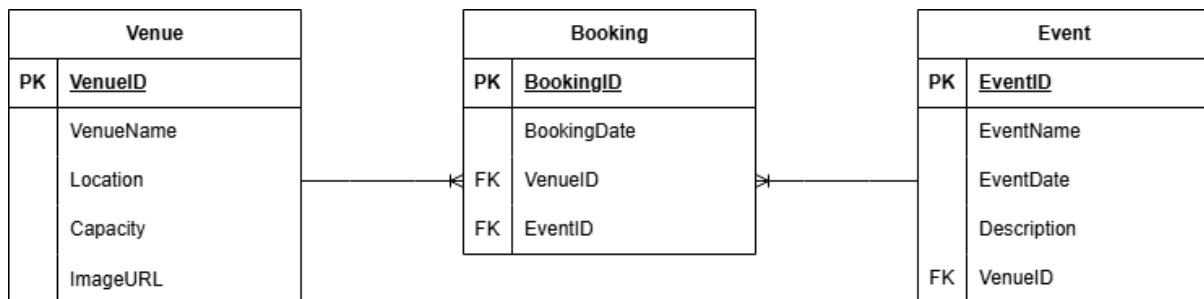
Part 1: Event Ease Report

Live Application URL: [EventEase-Koketso](#)

GitHub Repository Link: [EventEase](#)

A. Database Design

ERD Diagram



Database Script

```
IF OBJECT_ID(N'[__EFMigrationsHistory]') IS NULL
BEGIN
    CREATE TABLE [__EFMigrationsHistory] (
        [MigrationId] nvarchar(150) NOT NULL,
        [ProductVersion] nvarchar(32) NOT NULL,
        CONSTRAINT [PK__EFMigrationsHistory] PRIMARY KEY ([MigrationId])
    );
END;
GO

BEGIN TRANSACTION;
GO

CREATE TABLE [Venues] (
    [VenueId] int NOT NULL IDENTITY,
    [VenueName] nvarchar(max) NOT NULL,
    [Location] nvarchar(max) NOT NULL,
    [Capacity] int NOT NULL,
    [ImageURL] nvarchar(max) NOT NULL,
    CONSTRAINT [PK_Venues] PRIMARY KEY ([VenueId])
);
GO

CREATE TABLE [Events] (
    [EventId] int NOT NULL IDENTITY,
```

```

[EventName] nvarchar(max) NOT NULL,
[EventDate] datetime2 NOT NULL,
[Description] nvarchar(max) NOT NULL,
[VenueId] int NOT NULL,
CONSTRAINT [PK_Events] PRIMARY KEY ([EventId]),
CONSTRAINT [FK_Events_Venues_VenueId] FOREIGN KEY ([VenueId]) REFERENCES [Venues]
([VenueId]) ON DELETE CASCADE
);
GO

CREATE TABLE [Bookings] (
[BookingId] int NOT NULL IDENTITY,
[EventId] int NOT NULL,
[VenueId] int NOT NULL,
[BookingDate] datetime2 NOT NULL,
CONSTRAINT [PK_Bookings] PRIMARY KEY ([BookingId]),
CONSTRAINT [FK_Bookings_Events_EventId] FOREIGN KEY ([EventId]) REFERENCES [Events]
([EventId]) ON DELETE NO ACTION,
CONSTRAINT [FK_Bookings_Venues_VenueId] FOREIGN KEY ([VenueId]) REFERENCES [Venues]
([VenueId]) ON DELETE NO ACTION
);
GO

CREATE INDEX [IX_Bookings_EventId] ON [Bookings] ([EventId]);
GO

CREATE INDEX [IX_Bookings_VenueId] ON [Bookings] ([VenueId]);
GO

CREATE INDEX [IX_Events_VenueId] ON [Events] ([VenueId]);
GO

INSERT INTO [__EFMigrationsHistory] ([MigrationId], [ProductVersion])
VALUES (N'20260316024611_FixCascadePaths', N'8.0.25');
GO

COMMIT;
GO

```

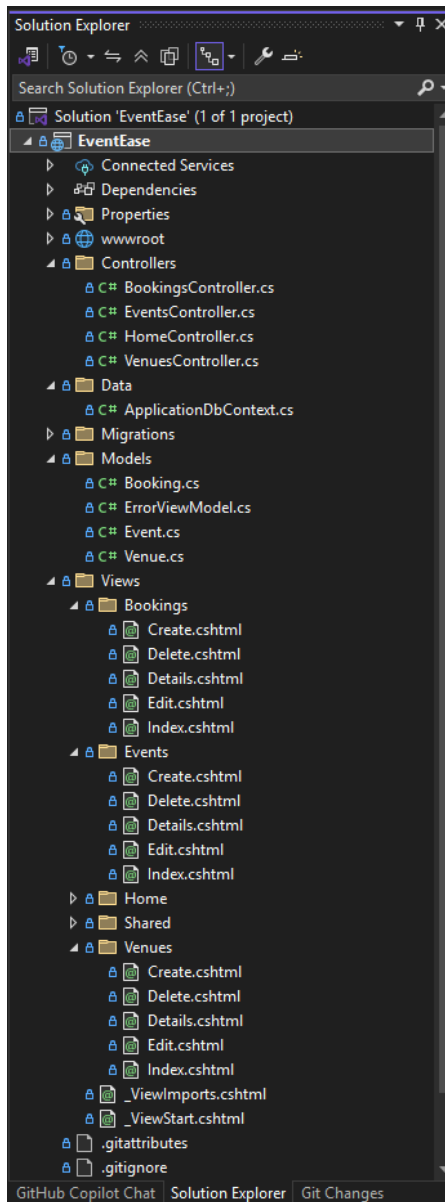
B. Web Application

The application was developed locally using Entity Framework Core and LocalDB for rapid prototyping.

Relevant Models, Views and Controllers for the following:

1. Venue

2. Event
3. Booking



The application supports full CRUD (Create, Read, Update, Delete) operations for Venues, Events, and Bookings.

```
1 using Microsoft.EntityFrameworkCore;
2 using EventEase.Data;
3 var builder = WebApplication.CreateBuilder(args);
4
5 // Add services to the container.
6 builder.Services.AddControllersWithViews();
7
8 // ADD THIS PART:
9 builder.Services.AddDbContext<ApplicationDbContext> (options =>
10     options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnection")));
11
12 var app = builder.Build();
13
14 // Configure the HTTP request pipeline.
15 if (app.Environment.IsDevelopment())
16 {
17     app.UseExceptionHandler("/Home/Error");
18     // The default HSTS value is 30 days. You may want to change this for production scenarios, see https://aka.ms/aspnetcorehsts
19     app.UseHsts();
20 }
21
22 app.UseHttpsRedirection();
23 app.UseStaticFiles();
24
25 app.UseRouting();
26
27 app.UseAuthorization();
28
29 app.MapControllerRoute(
30     name: "default",
31     pattern: "{controller=Home}/{action=Index}/{id?}");
32
33 app.Run();
```

Index

[Create New](#)

VenueName	Location	Capacity	ImageURL	
Dimension Data	Sandton	250		Edit Details Delete

EventEase Home Privacy Venues Events Bookings

Index

[Create New](#)

Event Name	Event Date	Description	Venue	
Takealot House of Beauty	3/17/2026	Takealot House of Beauty event is a show stopper in cosmetic shopping.	1	Edit Details Delete

EventEase Home Privacy Venues Events Bookings

Index

[Create New](#)

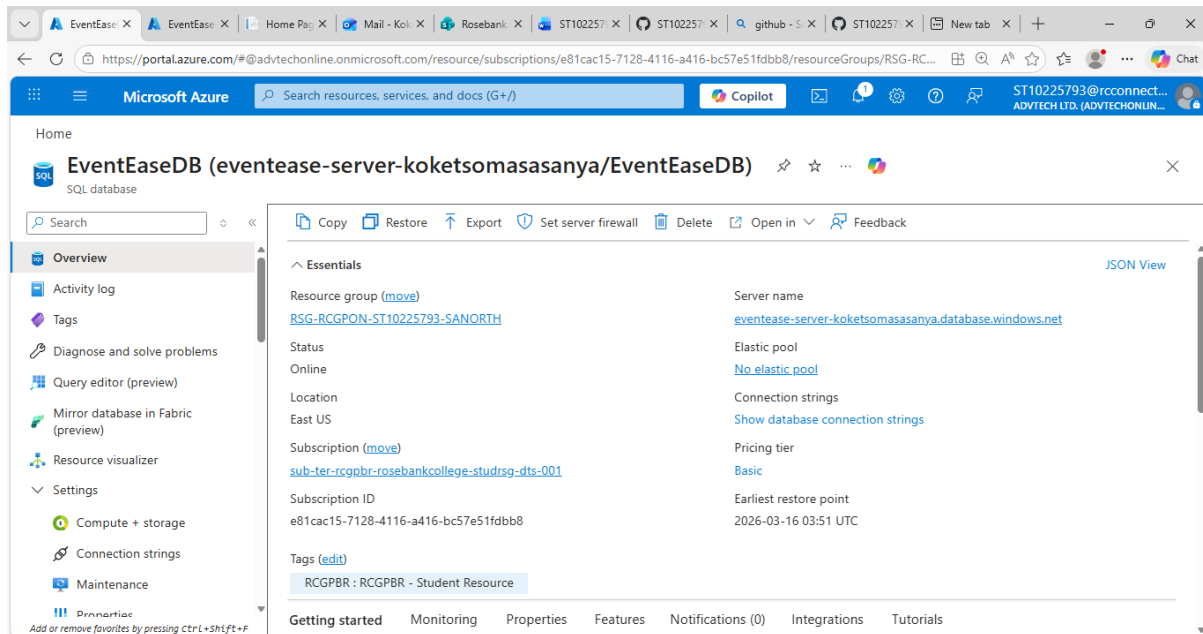
BookingDate	Event	Venue	
3/17/2026 5:00:00 PM	1	1	Edit Details Delete

C. Cloud Deployment

This section documents the migration of the EventEase application from a local environment to the Microsoft Azure Cloud.

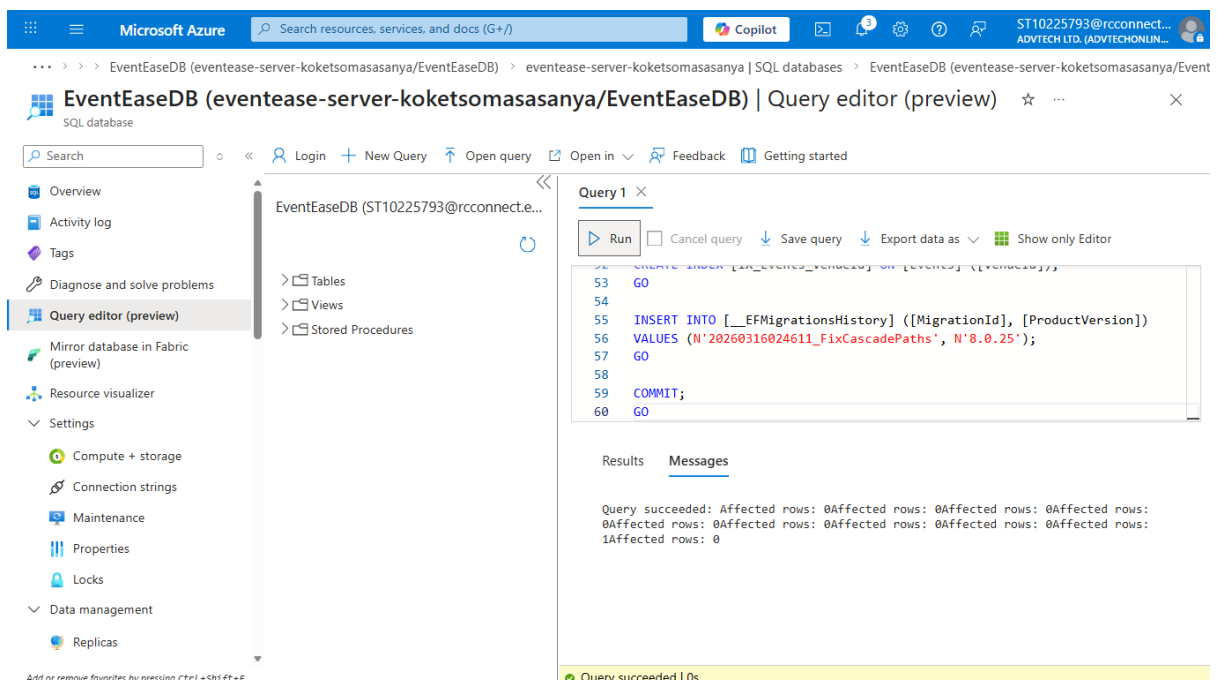
Azure SQL Database Provisioning

A PaaS SQL Database was provisioned using the Basic Tier to ensure cost-efficiency and compliance with institutional quotas.



Database Schema Deployment

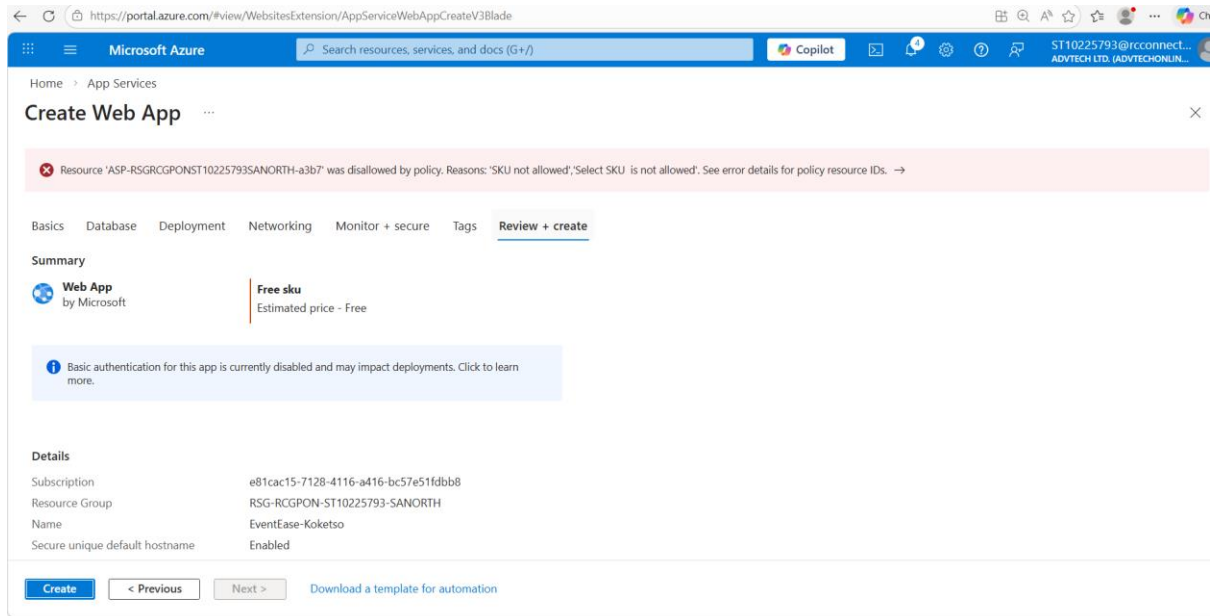
The schema was deployed to the cloud using SQL scripts generated via Entity Framework Migrations.



App Service Configuration & Security

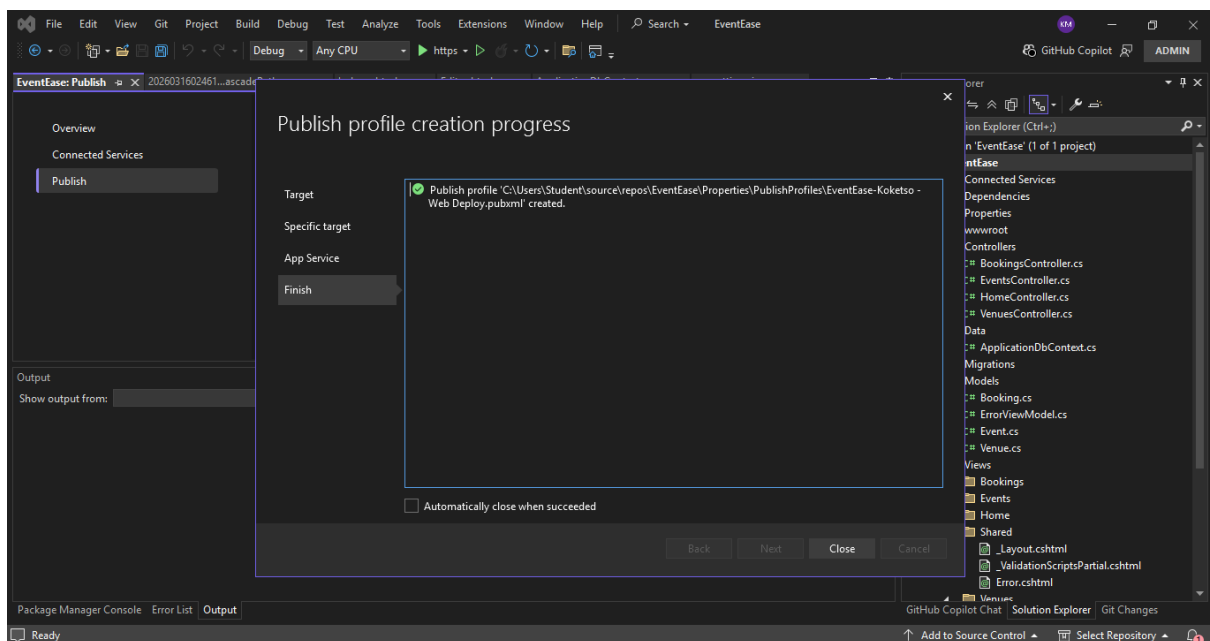
The web application was hosted on an Azure App Service (Basic B1 Plan). To ensure modern security standards, Microsoft Entra Managed Identity was implemented for a passwordless connection between the app and the database.

Challenge: Institutional policies initially blocked "Free" SKUs and public firewall access.



Resolution: Scaled to "Basic" tier and configured "Allow Azure Services" firewall rules to enable seamless internal communication.

Successful Publishing on VS



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Index

[Create New](#)

VenueName	Location	Capacity	ImageURL	
Azure Cloud Lab	Johannesburg	500		Edit Details Delete

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eventease-serv... EventEase-Kok... Details - EventE... Microsoft Azure... Mail - Koketso... beautiful office... beautiful office... beautiful-office... Summarize

Details

Venue

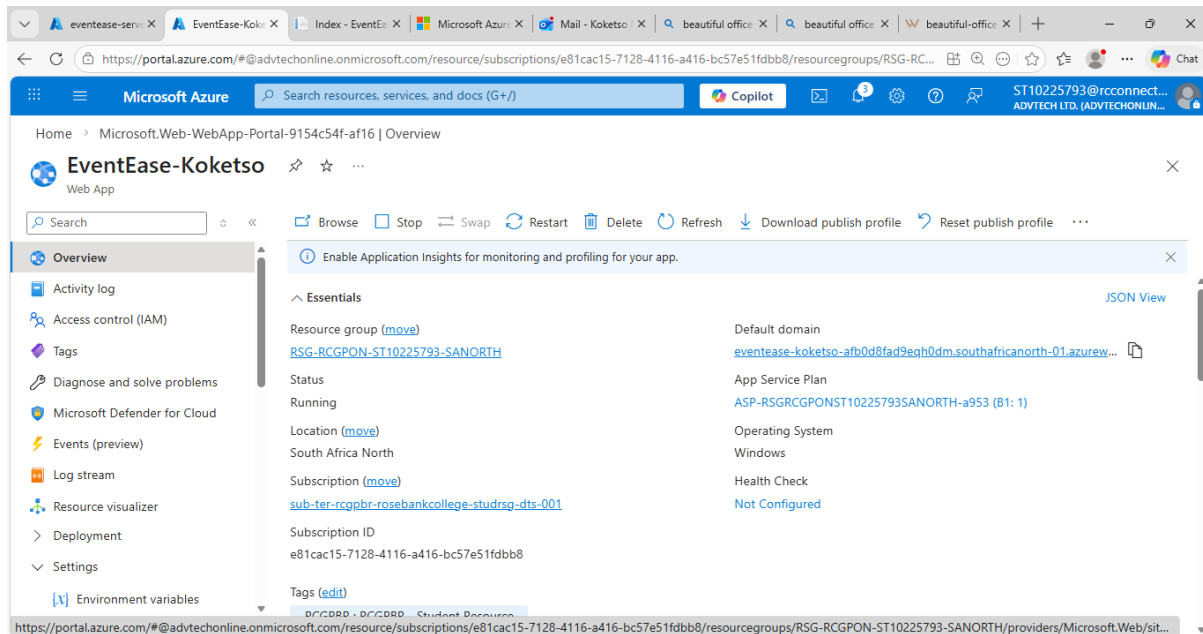
VenueName	Azure Cloud Lab
Location	Johannesburg
Capacity	500
ImageURL	

[Edit](#) | [Back to List](#)

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>

Azure SQL Database Setup



D. Cloud Theory Discussion

1. On-Premises vs. Cloud Computing

The transition of the EventEase application from a local development environment to Microsoft Azure represents a shift from decentralized hosting to a centralized, managed cloud infrastructure. According to (Mell & Grance, 2011), cloud computing provides "ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources."

Moving EventEase to Azure provided three distinct advantages:

- **Global Accessibility:** Unlike local hosting, which is restricted to the host machine's intranet, Azure provides a public endpoint, ensuring the application is accessible from any geographical location (Verma A, 2015).
- **Enhanced Security:** While LocalDB utilizes Windows Authentication, Azure Cloud security leverages **Microsoft Entra ID** and **Managed Identities**. This ensures a "Zero Trust" environment where resources are only accessible via verified service principals rather than stored connection strings (Microsoft, 2024).
- **High Availability and Reliability:** Azure SQL Database offers a built-in availability SLA of 99.99%, which is nearly impossible to achieve in an on-premises setup without redundant, expensive hardware (Chappell, 2019).

2. Service Model: Platform as a Service (PaaS)

EventEase was deployed using the **Platform as a Service (PaaS)** model through Azure App Service and Azure SQL Database.

Justification: The PaaS model was selected to maximize development efficiency. Under the **Shared Responsibility Model**, Microsoft manages the physical hardware, virtualization layer, and Operating

System (OS) patching, allowing the developer to focus exclusively on the application logic and data schema (Microsoft, 2023). This eliminates the "Administrative Overhead" associated with **Infrastructure as a Service (IaaS)**, where the developer would be responsible for configuring and securing the underlying Virtual Machine.

3. Scalability and Elasticity

The implementation of the **Basic B1 App Service Plan** provides a strategic foundation for both vertical and horizontal scaling.

- **Vertical Scaling (Scale Up):** The ability to increase the power of the existing instance (CPU/RAM) to handle complex processing tasks.
- **Horizontal Scaling (Scale Out):** The ability to add more instances of the web app to distribute high traffic loads (Bondi, 2014).

This elasticity ensures that the EventEase system remains responsive during peak event-booking seasons by dynamically adjusting resources to meet user demand, a core tenet of cloud economics.

Bibliography

Bondi, A., 2014. *Foundations of Software and System Performance Engineering*. Upper Saddle River, NY: Addison-Wesley.

Chappell, D., 2019. *A Technical Overview of Azure SQL Database*, s.l.: Microsoft Azure Whitepapers.

Mell, P. & Grance, T., 2011. *The NIST Definition of Cloud Computing.*, Gaithersburg, MD: National Institute of Standards and Technology (NIST).

Microsoft, 2023. *Shared responsibility in the cloud*. [Online]

Available at:

<https://eur03.safelinks.protection.outlook.com/?url=https%3A%2F%2Flearn.microsoft.com%2Fen-us%2Fazure%2Fsecurity%2Ffundamentals%2Fshared-responsibility&data=05%7C02%7CST10225793%40rconnect.edu.za%7Cd16a9bd829e64c44a7a908de83c568db%7Ce10c8f44f469448fbc0dd>

[Accessed March 2026].

Microsoft, 2024. *What are managed identities for Azure resources*. [Online]

Available at:

<https://eur03.safelinks.protection.outlook.com/?url=https%3A%2F%2Flearn.microsoft.com%2Fen-us%2Fentra%2Fidentity%2Fmanaged-identities-azure-resources%2Foverview&data=05%7C02%7CST10225793%40rconnect.edu.za%7Cd16a9bd829e64c44a7a908de83c568db%7Ce10c8f44f469>

[Accessed March 2026].

V. A., 2015. Cloud Computing: Concepts, Architecture and Challenges. *International Journal of Computer Applications*, 120(10).